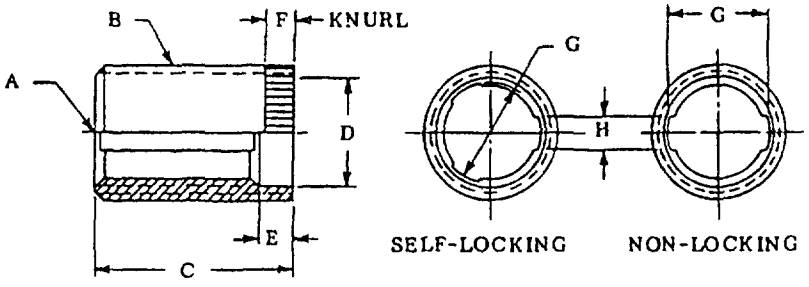
Metric
DOD-I-63274/1A
14 February 1983
SUPERSEDING
DOD-I-63274/1
26 September 1977

MILITARY SPECIFICATION SHEET

INSERT, SCREW THREAD-THIN WALL, LOCKED IN, METRIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the insert described herein shall consist of this specification and the latest issue of specification DOD-I-63274



NOTES:

1. Unless otherwise specified, dimensions are in millimeters, and shall apply after plating and before the addition of solid film lubricant.
2. The centerline of the internal thread locking feature shall be approximately mid-length of internal thread.

TABLE I: Part dimensions and dash numbers

DASH NUMBER *			A INTERNAL THREAD	B EXT THREAD ALTERED MINOR DIA		C LENGTH	D DIA	E	F	G DIA	H -	MIN SHEAR ENGAGEMENT AREA (mm^2) (REQT 5)
17-4PH CRES	ALLOY STEEL	A-286 CRES		THREAD SIZE	MAX MINOR DIA	±0 3	±0.20	±0 20	±0 3	MIN	MIN	
1L	1AL	1CL	MJ3x0.5	M5x0 8	4 044	5 1	3 17	2 15	1 6	2.84	0 90	25 44
2	2A	2C								2 86	1 20	
3L	3AL	3CL	MJ4x0.7	M6x1	4.954	6 9	4 21	2 55	1 9	3.72	1 15	49 73
4	4A	4C								3 74	1 50	
5L	5AL	5CL	MJ5x0.8	M7x1	5 973	7 6	5 23	2 95	2.3	4.56	1 25	63 32
6	6A	6C								4 69	1 70	
7L	7AL	-				9 1				4 56	1 25	87.28
8	8A	-								4 69	1 70	
9L	9AL	9CL	MJ6x1	M8x1	7 016	8 9	6 28	3 15	2 4	5.44	1 50	95.29
10	10A	10C								5.58	2.00	
11L	11AL	-				10 7				5 44	1.50	128.20
12	12A	-								5 58	2 00	
13L	13AL	13CL	MJ8x1 25	M10x1	9 056	12 8	8 32	3 95	2 9	7 28	2.00	198.10
14	14A	14C								7 58	2 70	
15L	15AL	-				15 7				7 28	2 00	264.70
16	16A	-								7 58	2 70	
17L	17AL	17CL	MJ10x1.5	M12x1	11 110	16 2	10 37	4 65	3 3	9.14	2 50	321 30
18	18A	18C								9 43	3 20	
19L	19AL	-				19 9				9 14	2 50	423.00
20	20A	-								9 43	3 20	
21L	21AL	21CL	MJ12x1 25	M14x1	13 116	19 9	12.37	4 65	3 5	11 14	2.50	478.20
22	22A	22C								11.43	3.20	
23L	23AL	-				24 8				11 14	2 50	632.50
24	24A	-								11 43	3 20	
25L	25AL	25CL	MJ14x1.5	M16x1	15 116	23 1	14.37	4 65	3.6	13 15	2 50	659.00
26	26A	26C								13 30	3.20	
27L	27AL	-				28 8				13 15	2 50	864.50
28	28A	-								13 30	3 20	

* "L" suffix indicates self-locking insert

FSC 5340

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:31Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

1. Material: Steel, alloy, composition 4130 (UNS G41300) per AMS 6370. Steel, corrosion-resisting, composition 17-4PH (UNS S17400) per AMS 5643. Steel, corrosion-resisting, composition A-286 (UNS K66286) per AMS 5731 or AMS 5734.

2. Protective finish or treatment: Alloy steel (UNS G41300) shall be cadmium plated plus solid film lubricant in accordance with general specification. Corrosion-resisting steel (UNS S17400) shall be solid film lubricant in accordance with general specification. Corrosion-resisting steel (UNS K66286) shall be silver plated in accordance with general specification.

3. Surface texture: 3.2 µm except knurling.

4. Threads: Threads shall be in accordance with Table I and general specification. Thread dimensions shall apply before application of solid film lubricant.

5. Shear engagement area: Shear engagement area is the assembled dimensional value for the overall engaged area of mating thread members. It does not represent a dimension of either of the members in an unassembled condition.

6. Military part number: Consists of the letter D, plus the basic number of the specification sheet, and a dash number taken from Table I.

Example of Military part number:

D63274/1 -	3CL
Basic number of specification sheet	Dash Number

Explanation of part number example:

Insert, Steel, CRES, A-286, MJ4x0.7 internal thread size, and a self-locking feature.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:31Z
Check the source to verify that this is the current version before use.

HOLE PREPARATION

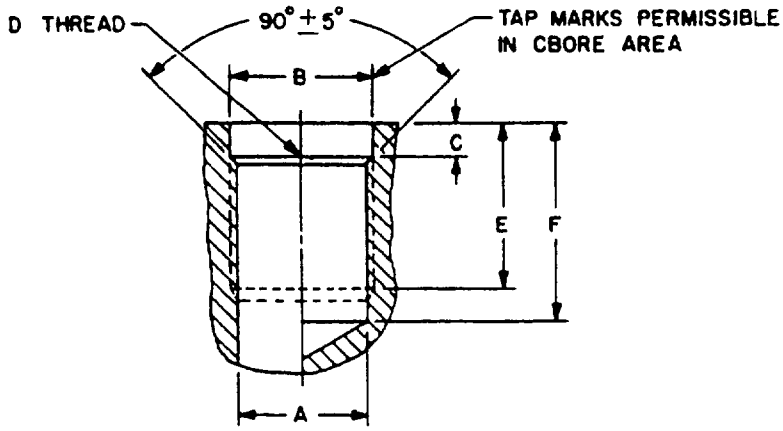


TABLE II. Installation criteria in parent material

INSERT DASH NUMBER REF	EXTERNAL THREAD SIZE OF INSERT REF	A TAP DRILL DIA		B CBORE DIA		C CBORE DEPTH +0.25 -0.00	D THREAD ALTERED MINOR DIA (NOTE 3)	E FULL THREAD DEPTH MIN	F DRILL DEPTH BLIND HOLE MIN	INSERT REMOVAL DRILL SIZE (NOTE 7)
		MAX	MIN	MAX	MIN					
1, 2	M5x0 8	4.24	4 14	5 10	5 00	1.70	M5x0.8	6 1	8 1	4 9
3, 4	M6x1	5 11	5 01	6 10	6.00	2.00	M6x1	7 9	10 4	5 8
5, 6	M7x1	6.13	6 03	7 10	7.00	2 40	M7x1	8 6	11.1	6 9
7, 8								10 1	12.6	
9,10	M8x1	7 17	7 07	8 10	8.00	2.50	M8x1	9 9	12.4	7.8
11,12								11 7	14.2	
13,14	M10x1	9.24	9 11	10.13	10.00	3.00	M10x1	13.8	16 3	9 9
15,16								16.7	19.2	
17,18	M12x1	11.29	11 16	12.13	12.00	3 40	M12x1	17 2	19.7	12 0-
19,20								20 9	23 4	
21,22	M14x1	13.32	13 17	14.15	14.00	3.60	M14x1	20.9	23 4	14 0
23,24								25.8	28.3	
25,26	M16x1	15 32	15 17	16 15	16 00	3 70	M16x1	24 1	26 6	16.0
27,28								29 8	32.3	

NOTES:
1. Diameter "A" and thread shall be concentric within 0.15 FIM
2. Axis of hole shall be normal to entry surface or provide spotface when required.
3. Threads "D" shall be class-4H5H.
4. Machined surfaces shall be 3.2 µm in accordance with ANSI B46.1.
5. All dimensions are in millimeters.
6. Install insert:
   a. These inserts are primarily designed for use in aluminum, magnesium and other nonferrous materials that do not exceed Brinell 187 (3000kg load and 10 mm ball). Use in corrosion-resisting steels, titanium and hardened ferrous materials will normally require broaching serrations in counterbore to accept the insert knurls during swaging operation.
   b. Use of insert manufacturer's wrench and swaging tool is mandatory.(Rosan, Inc. Newport Beach, CA - FSCM 83324)
   c. Install inserts into hole until the top of insert is 0.40-0.65 below boss surface.
   d. Place swage tool in insert and apply a downward force sufficient to bottom the tool shoulder against the boss surface which will effect full swageout and external lock setting.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:31Z
Check the source to verify that this is the current version before use.

7. Replacements of inserts are made with same size inserts as those removed. Using removal drill size shown in Table II, drill to depth "C" +0.6, then back out insert using installation wrench or a square type screw extractor. Remove loose chips, reinspect hole and then install new insert per note 6.

Custodians:
Army - AR
Navy - OS
Air Force - 99

Review activities:
Army - AV, MI
Air Force - 11
DLA - IS

User activities:
Army - ME, AT, CR
Navy - MC

Preparing activity:
Army - AR

(Project 5340-1462)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:31Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

BUSINESS REPLY CARD

FIRST CLASS PERMIT NO 12062 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

COMMANDER
US ARMY ARMAMENT RESEARCH & DEVELOPMENT
COMMAND
ATTN: DRDAR-TST-S
DOVER, NEW JERSEY 07801

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:31Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS This form is provided to solicit beneficial comments which may improve this document and enhance its use DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government Fold on lines on reverse side, staple in corner, and send to preparing activity Attach any pertinent data which may be of use in improving this document If there are additional papers, attach to form and place both in an envelope addressed to preparing activity A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed

NOTE This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements

DOCUMENT IDENTIFIER *(Number)* AND TITLE

DOD-I-63274//A INSERT, SCREW THREAD-THIN WALL, LOCKED IN, METRIC

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1 ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE LOOSF OR AMBIGUOUS? PLEASE EXPLAIN BELOW

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDED WORDING CHANGE

C REASON FOR RECOMMENDED CHANGE(S)

2 REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:31Z
Check the source to verify that this is the current version before use.